

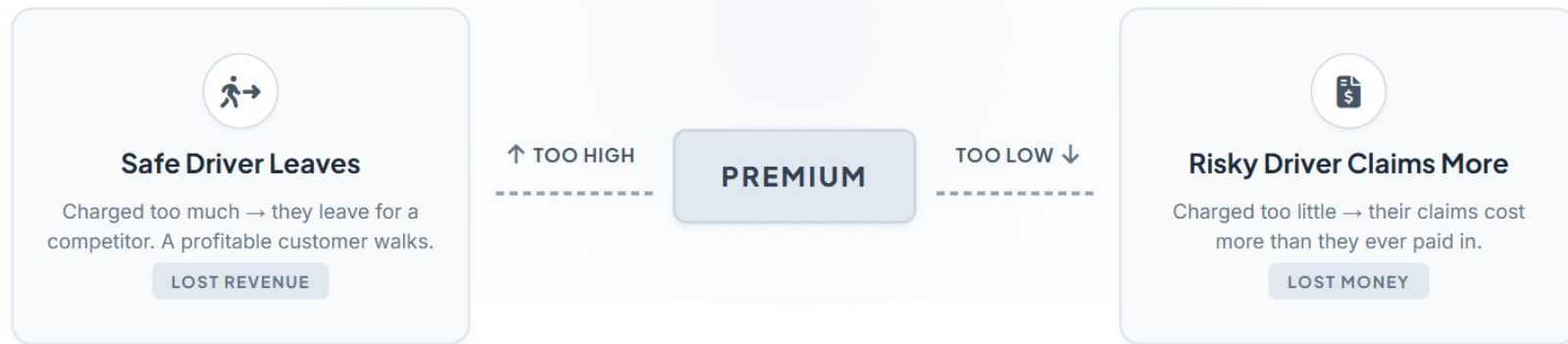
ML Project

Porto Seguro's Safe Driver Prediction

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Mispricing costs money



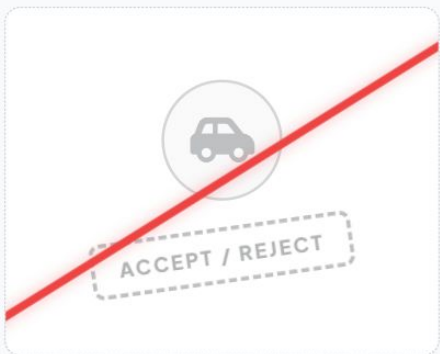
PORTO SEGURO FRAMING

"Inaccurate risk prediction makes insurance more expensive for good drivers."

Not predicting claims, but ranking risk

✗ NOT WHAT WE'RE DOING

Binary Accept / Reject



The model does **not** decide who to accept or reject as a binary pass/fail rule.

✓ WHAT WE'RE DOING

Risk Ranking & Ordering



Sorts drivers from lowest to highest risk. **Pricing uses the order.**

What we predict: Did this policyholder file a claim in the following year? (Yes / No)

What the business uses: The ordering. Not the probability.



96% accurate and completely useless

Predict "no claim" for every driver: 96.331% accurate.

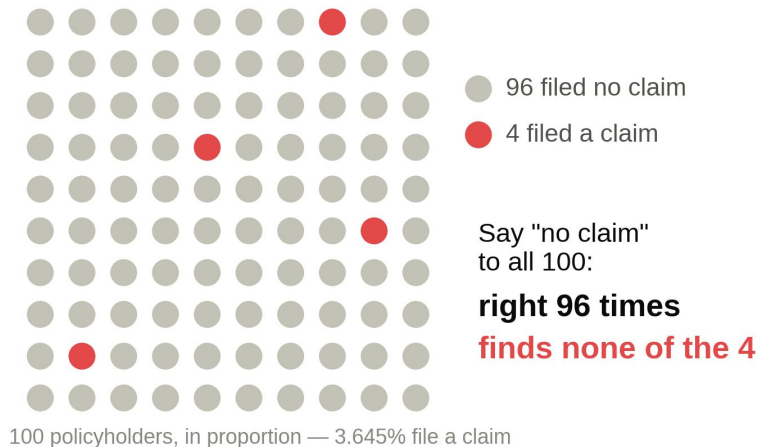
Finds zero of the claims. Only **3.645%** of policyholders claim.

What we measure instead: how well the model orders drivers by risk.

Random ordering = 0. Ours = **0.27**.

Not a percentage. The best published result on this dataset —

three months of work, stacked models — is 0.297.





What the data could and couldn't tell us

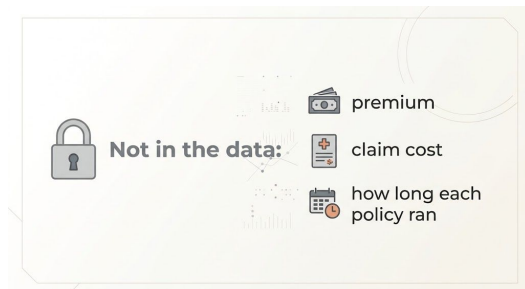
What the data could and couldn't tell us

595,212 policies · 57 fields · 21,694 claims (3.645%)

Blanks are stored as -1, not as blanks — so a standard "any missing data?" check reports a perfectly clean dataset.

13 fields affected: two of them 69% and 45% blank, the rest under 2%.

Every field name is anonymized: We can tell whether a field is about the driver, the region or the vehicle. We cannot tell what it measures.



The 20% nobody has looked at

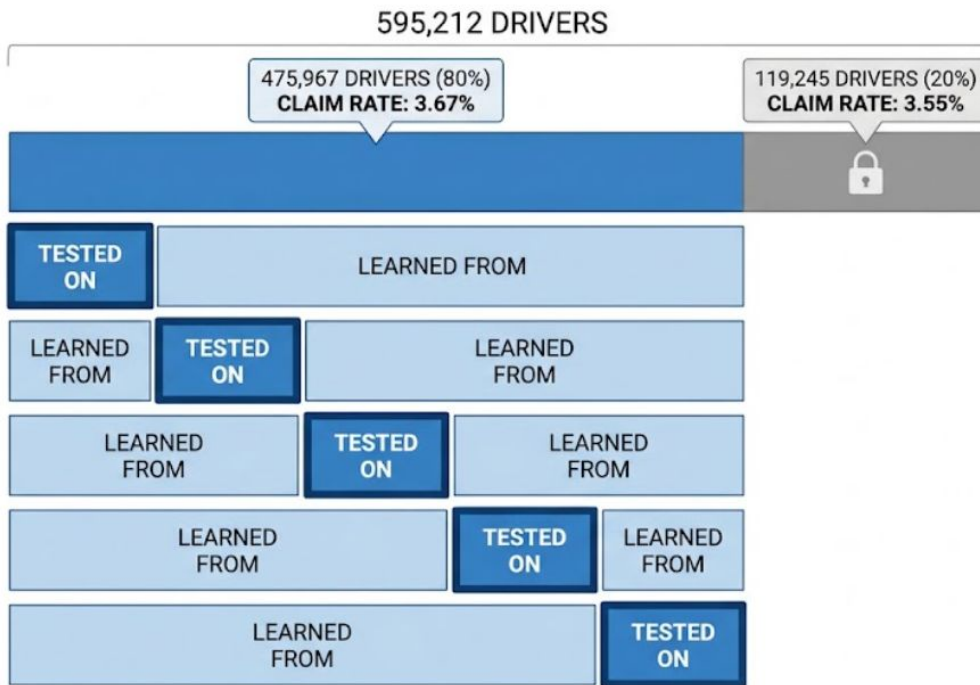


Inside the 80%:

We test on drivers the model has never seen, five different ways, and report the average and spread.

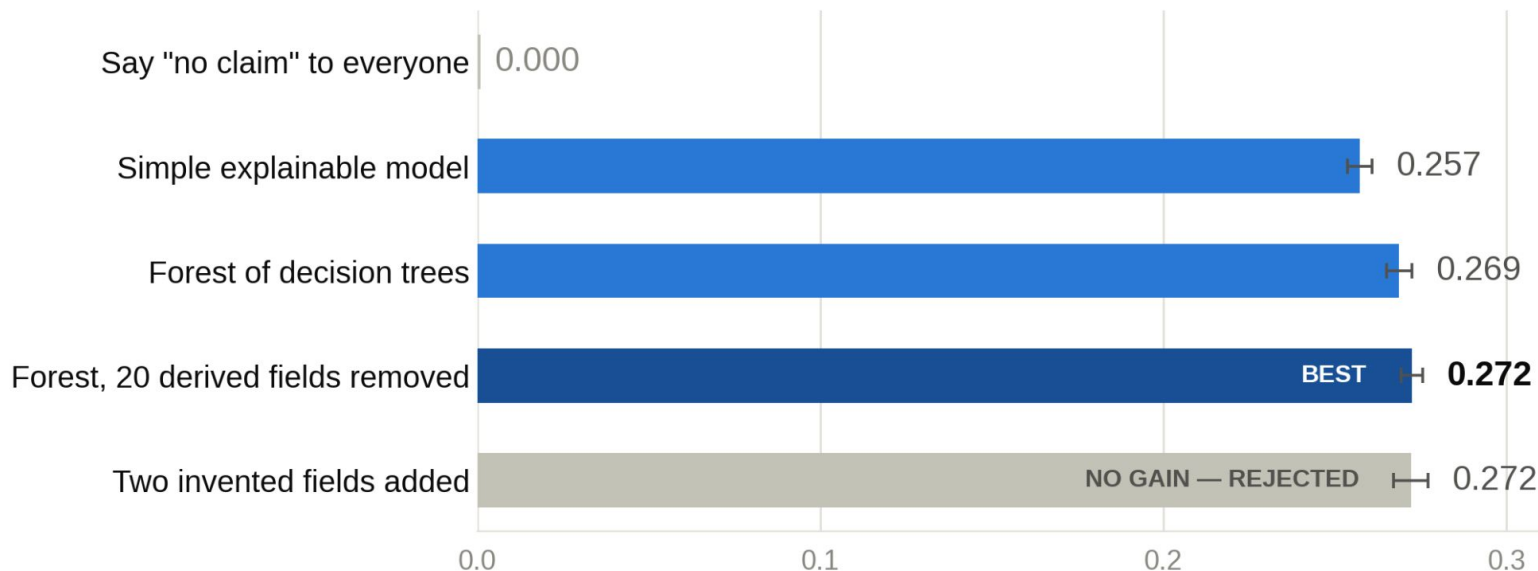


Which pile a driver lands in is decided by a fingerprint of their ID — so the **split is identical on any laptop**, in any run, even if the file gets reordered.



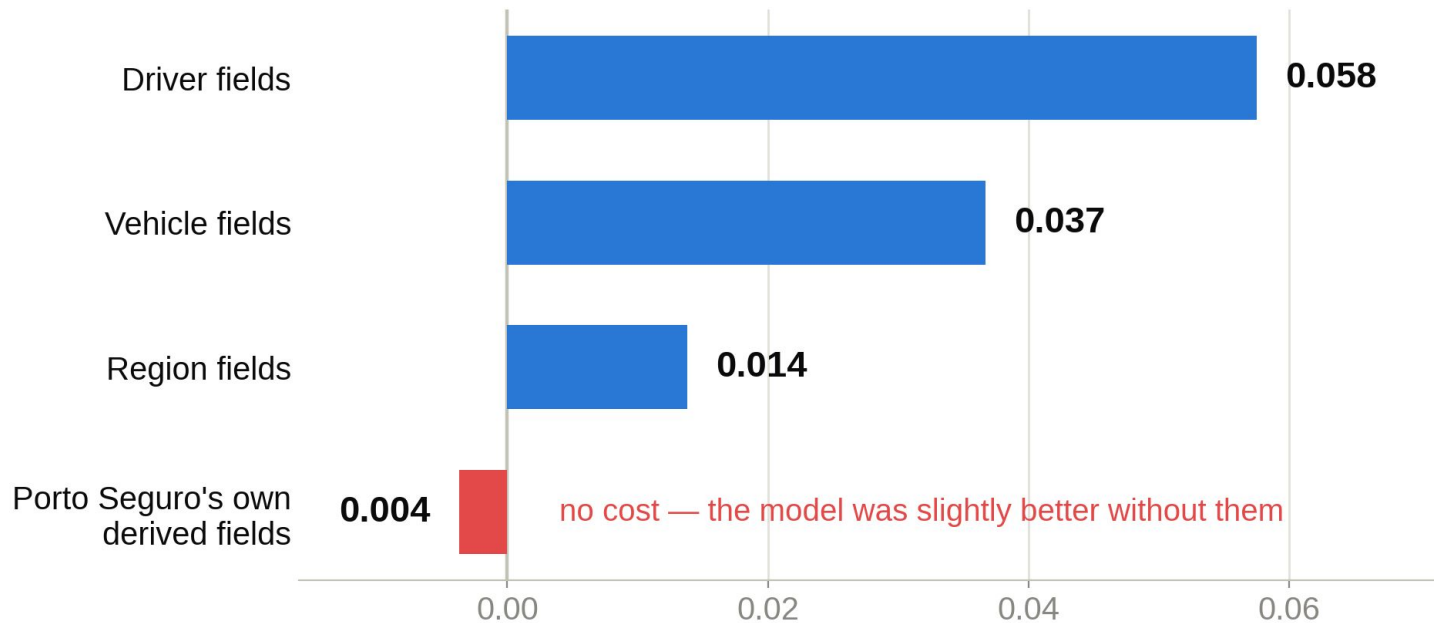


What we built, in order



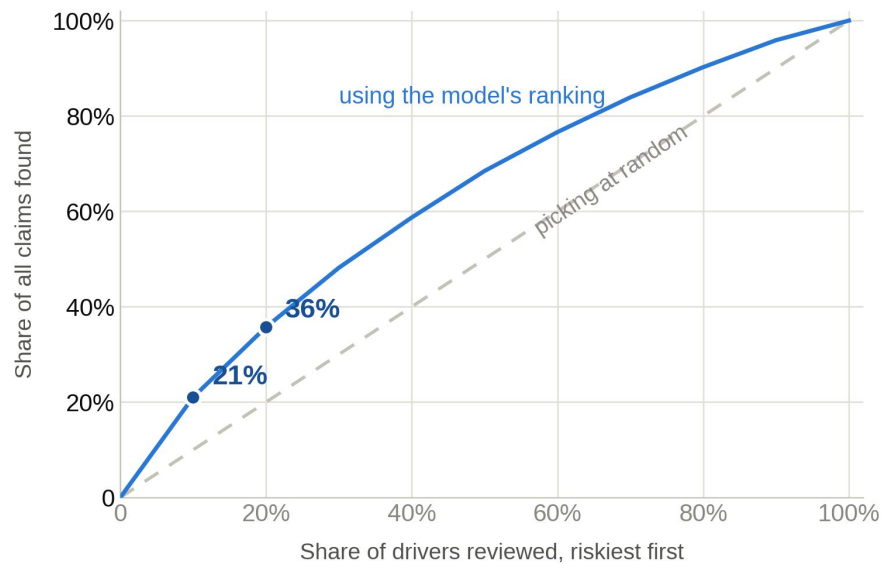
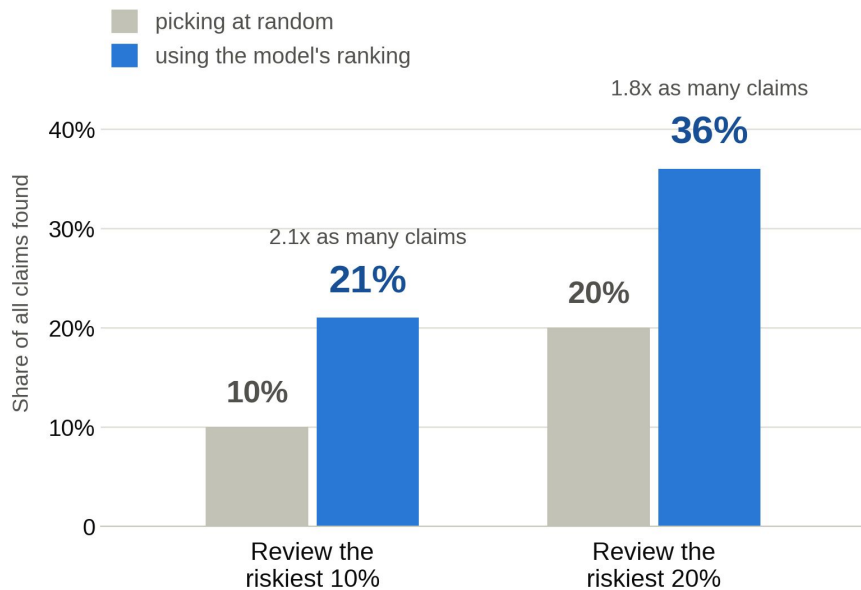


What the model found



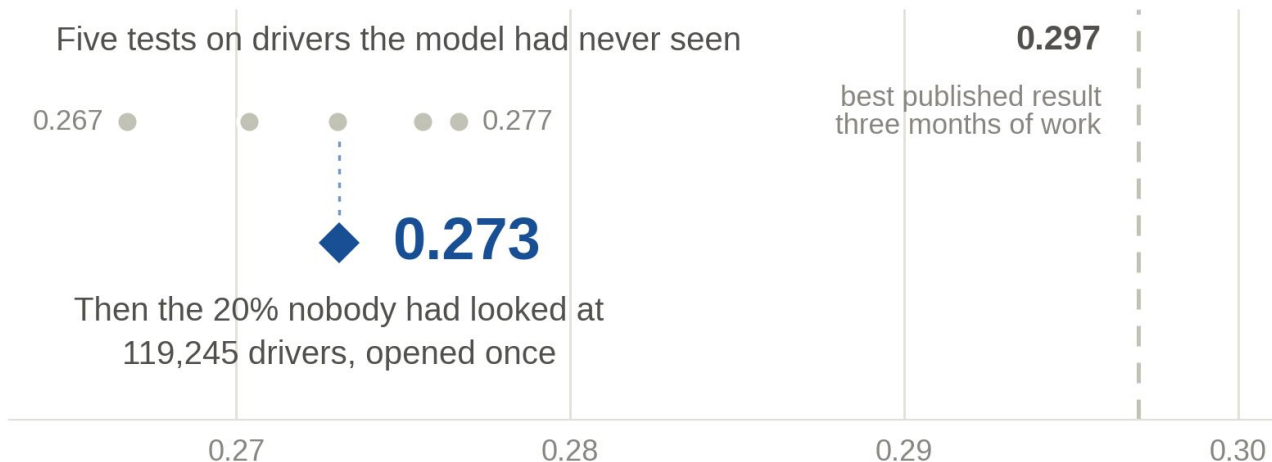


What it's worth





Where we stopped, and why



WHAT WE KNOW IS STILL WRONG

5,809 drivers on the field the model leans on most. We price them too low.





What we'd do next, and what we'd need

NEXT, IN ORDER OF EXPECTED PAYOFF



1 Fix the under-pricing

Handle small groups properly instead of averaging them into safer neighbours.

measured

low effort



2 A bounded round of tuning

Time budget agreed before it starts, not after.

untested

time-boxed



3 Combine the two models

Use the explainable model and the forest together rather than choosing between them.

untested

larger job



WHAT WE'D ASK THE BUSINESS FOR

Documentation of what the fields actually mean.

Without it we cannot name a risk driver, explain a price to a customer, or check the model for unfair bias.

That last one blocks deployment.